Grade 8 | Name____________________
Unit 4 | Date____________________
Lesson 11 Practice | Period__________________

1. Select all the expressions that are equivalent to the expression $m^{-3}k^5 \cdot mk^{-1}$

- ☐ $\frac{k^5m}{km^3}$
- ☐ $\frac{k^4}{m}$
- ☐ $\frac{1}{m^2k^4}$
- ☐ $m^{-2} \cdot k^2$
- ☐ $m^{-2}k^4$

2. Select all the expressions that are equivalent to the expression $(3a^{-4}b^{-3})^3 \cdot (-3a)^{-2}$

- ☐ $\frac{27a^{-12}b^{-9}}{9a}$
- ☐ $\frac{3a^{-13}b^{-9}}{a}$
- ☐ $\frac{27a^{-14}}{3b^6}$
- ☐ $\frac{3a^{-7}}{27a^7b^{-2}}$
- ☐ $(3a^{-4}b^{-1})^3$

For questions 3–10, generate an equivalent expression for each by applying the laws of exponents. Use positive exponents and the fewest factors.

3. $x^5 \cdot (x^3)^{-4}$

4. $\frac{(x^7)^3}{x^{-7}}$

5. $(-4x)^{-2} \cdot (-x^3)^2$

6. $\frac{(6xb^{-2})^2}{(2x^{-3}b^{-4})^{-3}}$

7. $x^{-7} \cdot x^2 \cdot (x^3)^{-4}$

8. $\frac{3x^{-4}y^6}{-18x^3y}$

9. $\left(\frac{x^4}{3}\right)^{-3} \cdot \frac{x^3}{(4x^5)^{-2}}$

10. $\left(\frac{-2x^0y^4}{5x^{-3}y}\right)^{-2}$